



HEALTHCARE DATASET

Healthcare Analytics Dashboard • Report

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DATA ANALYTICS

HEALTHCARE

DASHBOARD REPORT

The **Healthcare Analytics Dashboard** is a data-driven visual report built to summarize and interpret large-scale hospital and patient records. Healthcare organizations generate enormous volumes of data every day from admissions, billing, treatments, and patient demographics. Without proper analysis, this data holds little value. This dashboard converts raw healthcare records into clear, actionable insights using Key Performance Indicators (KPIs) and interactive visual charts.

This report is based on a healthcare dataset containing patient billing, admission, and demographic information. The dashboard is organized into eight core analytical components — four summary KPI metrics and four visual breakdowns — which together provide a 360-degree view of hospital operations, financial performance, and patient distribution.

PURPOSE

To analyze hospital billing, admissions, and patient trends for better healthcare decision-making.

SCOPE

Covers billing amounts, admission types, medical conditions, patient age groups, and yearly trends.

The eight datasets analyzed in this report are:

1 Total Billing Amount

3 Total Patients

5 Billing Amount by Medical Condition

7 Patients by Age Group

2 Total Admissions

4 Average Billing per Patient

6 Admission Count by Type

8 Admissions Over Time

Total Billing Amount

\$

1.28 Billion

TOTAL BILLING AMOUNT

The **Total Billing Amount** represents the cumulative revenue generated across all patient records in the dataset, amounting to approximately **1.28 billion** in currency value. This figure aggregates every invoice raised for treatments, procedures, hospital stays, and medical services across all recorded medical conditions and admission types.

As a top-level financial KPI, this metric gives hospital administrators an immediate sense of overall financial scale. It acts as the anchor figure against which all other billing-related breakdowns — such as billing by medical condition — can be compared and validated.

- Reflects total revenue across the entire patient base
- Used as a baseline for financial planning and budgeting
- Helps identify overall organizational financial health

Total Admissions

A

56K

TOTAL ADMISSIONS

The **Total Admissions** metric counts every instance a patient was admitted to the hospital, totaling around **56,000** admission records. Since a single patient may be admitted multiple times for different conditions, this number is naturally higher than the total unique patient count.

This KPI is essential for understanding hospital capacity utilization, staffing needs, and resource allocation. Tracking admissions volume helps administrators anticipate bed occupancy, plan for peak periods, and evaluate the effectiveness of patient flow management across elective, urgent, and emergency care.

- Measures total patient intake events, not unique patients
- Supports hospital capacity and staffing planning
- Feeds directly into the Admission Type and Admissions Over Time analyses

Total Patients

P

40K

TOTAL PATIENTS

The **Total Patients** KPI reports the count of distinct individuals present in the dataset — roughly **40,000 unique patients**. This figure is lower than total admissions because many patients return for repeat visits, follow-ups, or treatment of multiple medical conditions over time.

Understanding the unique patient base is critical for population health management. It allows the hospital to estimate how many distinct people it serves, supports demographic segmentation (such as the age-group breakdown), and helps in calculating per-patient metrics like average billing.

- Counts unique patients, independent of visit frequency
- Basis for demographic and age-group analysis
- Used to calculate per-capita financial metrics

Average Billing per Patient

 Σ **25.55K**

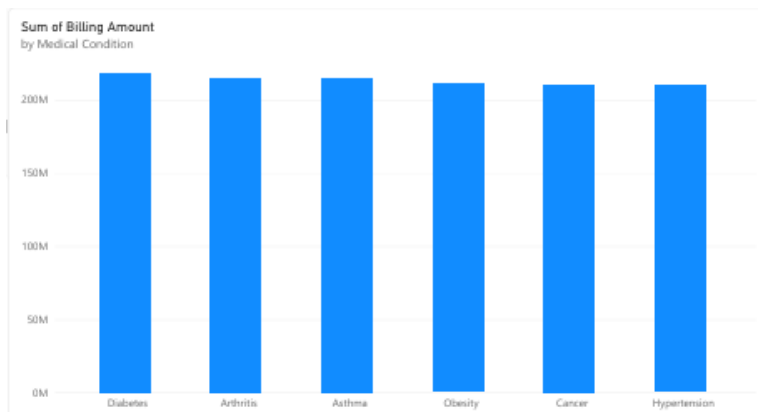
AVERAGE BILLING

The **Average Billing** figure, approximately **25,550**, is derived by dividing the total billing amount by the number of admissions (or patients). It represents the typical cost incurred per hospital visit or treatment episode.

This normalized metric is more useful than the raw total billing figure for comparing cost efficiency across departments, medical conditions, or time periods, since it accounts for volume differences. A rising average billing trend may indicate increasing treatment costs, higher case complexity, or inflation in medical service pricing.

- Calculated as Total Billing divided by Number of Admissions
- Useful for cost-efficiency comparisons
- Acts as an early indicator of rising treatment costs

Billing by Medical Condition

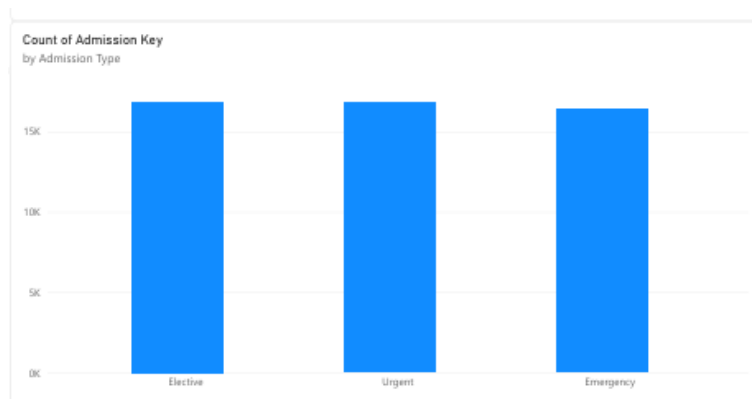


SUM OF BILLING AMOUNT — BY MEDICAL CONDITION

This bar chart breaks down total billing across six major medical conditions: **Diabetes**, **Arthritis**, **Asthma**, **Obesity**, **Cancer**, and **Hypertension**. Each bar reflects the cumulative billing amount associated with patients diagnosed under that condition, with values clustering close to the 200-220 million range for every category.

- Diabetes and Arthritis show marginally higher billing totals
- All six conditions contribute a broadly similar share of revenue
- Supports department-wise budget and resource planning

Admission Count by Type



COUNT OF ADMISSIONS — BY ADMISSION TYPE

Hospital admissions are classified into three categories: **Elective**, **Urgent**, and **Emergency**. The chart shows that Elective admissions are the most frequent, followed closely by Urgent and then Emergency admissions, with each category recording roughly 15,000-17,000 cases.

- Elective admissions form the largest share of total cases
- Urgent and Emergency admissions remain closely matched
- Helps balance planned vs. reactive hospital resource allocation

Patients by Age Group



PATIENTS BY AGE GROUP (TOTAL: 40.235K)

This donut chart segments the total patient population of **40,235** into distinct age groups. The largest segments dominate the chart, indicating that the hospital serves a substantial share of middle-aged and senior patients.

- Total patient base of 40,235 segmented into age bands
- Middle-aged and senior groups make up the largest proportion
- Guides age-specific care programs and specialty staffing

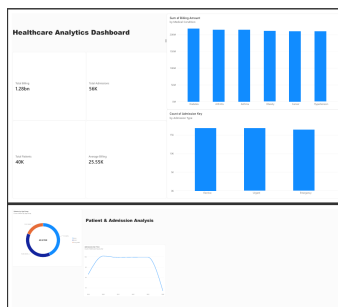
Admissions Over Time



COUNT OF ADMISSIONS — BY YEAR

This line chart tracks the trend of hospital admissions across successive years. Admission volumes rise sharply in the early years, reach a stable plateau through the middle years, and then show a noticeable decline toward the most recent year in the dataset.

- Admissions grow steadily before stabilizing across middle years
- A visible dip appears in the most recent year of records
- Useful for forecasting and identifying year-over-year patterns



HEALTHCARE ANALYTICS DASHBOARD — FULL OVERVIEW

Key Takeaways

The Healthcare Analytics Dashboard successfully consolidates complex hospital data into eight clear, decision-ready insights. With a total billing of **1.28 billion** generated from **56K admissions** across **40K unique patients**, and an average billing of **25.55K** per case, the hospital demonstrates a large, financially significant, and steadily operating patient base.

Billing is spread fairly evenly across six major medical conditions, admissions lean toward planned (elective) care while maintaining strong emergency readiness, the patient base skews toward middle-aged and senior groups, and admission volumes have remained largely stable over time with a recent dip worth further monitoring.

Overall, this dashboard equips hospital administrators, finance teams, and healthcare planners with the insights needed to optimize resource allocation, control costs, and improve patient care delivery. Future analysis could extend this study by correlating billing with treatment outcomes or exploring seasonal admission patterns in greater depth.